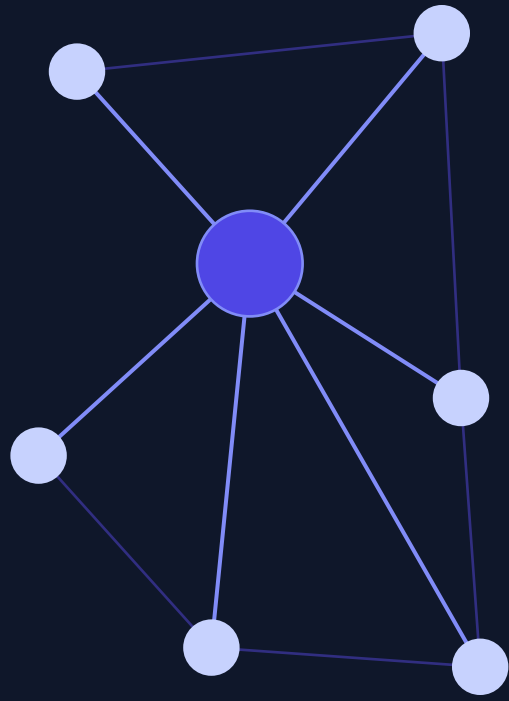
DEEP LEARNING ASSIGNMENT

TRANSFORMER NETWORKS



Presented by **S. Ashwin**

M.Sc. Computer Science (Data Analytics)

What is a Transformer?

- A deep learning architecture for sequential and contextual data
- Introduced in “Attention Is All You Need” (2017)
- Does not process tokens strictly one after another, unlike RNNs
- Core idea: the Attention Mechanism
- Now fundamental to NLP and Generative AI

2017

Attention Is All You Need

Transformers understand relationships between tokens using attention rather than recurrence.

Before Transformers

RNN

Recurrent Neural Network

One element at a time

Hidden state as memory

Weak on long-range links

LSTM

Long Short-Term Memory

Gates keep key information

Better on long sequences

Still sequential

GRU

Gated Recurrent Unit

Simpler than LSTM

Fewer parameters, faster

Still sequential

Transformer

Attention-based

Replaces recurrence

Parallel processing

Long-range context

Why Transformers?

1. Sequential processing slows training
2. Hard to parallelize computation
3. Distant words lose influence (vanishing gradients)
4. Difficult to scale to huge data and models

THE QUESTION

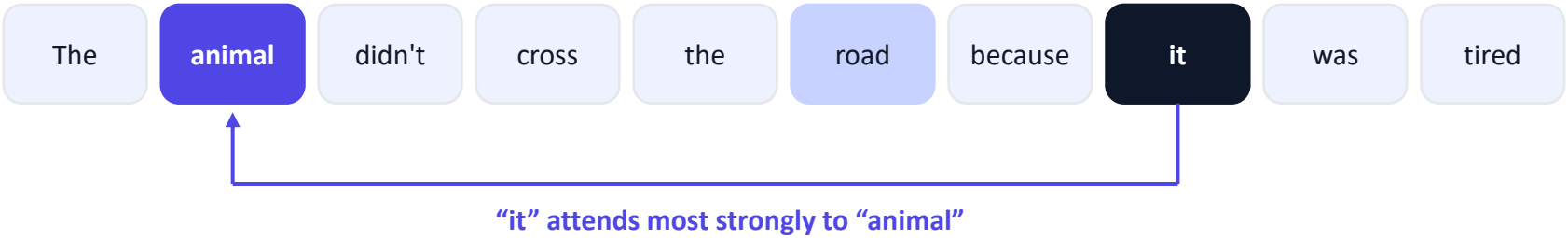
Can we understand relationships between all tokens without going one by one?

↓

Use Attention.

CORE CONCEPT

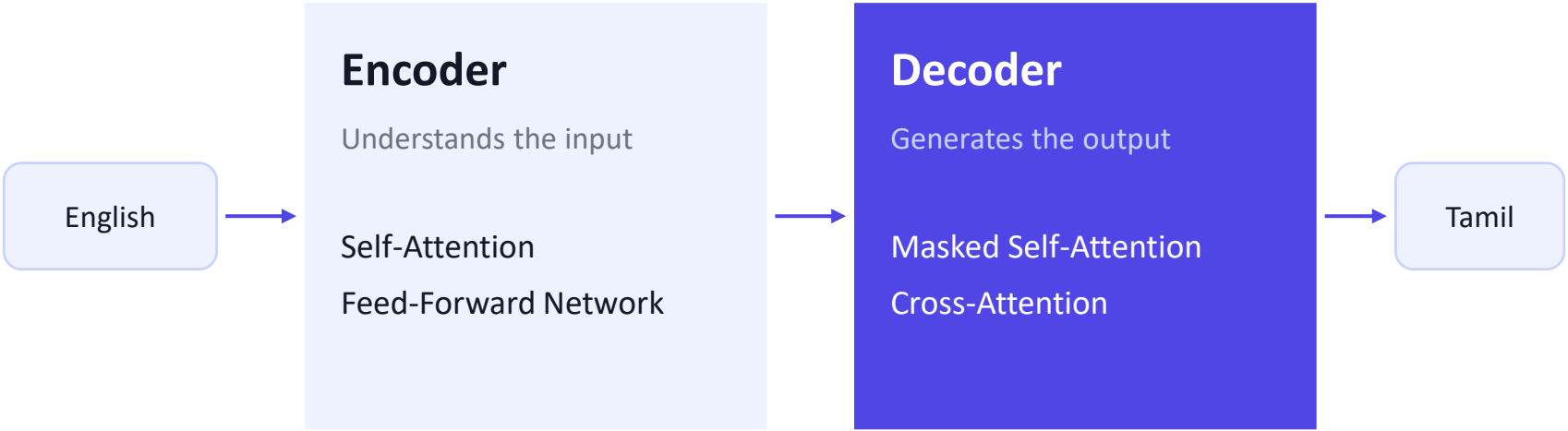
Attention Instead of Recurrence





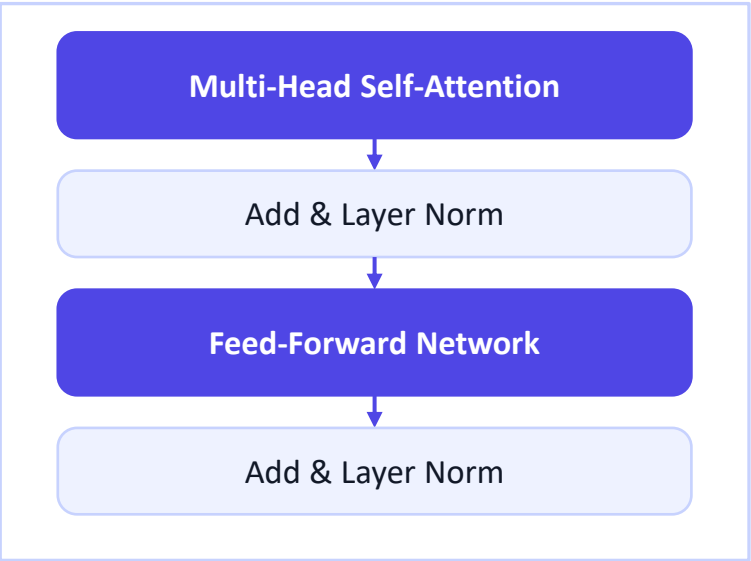
Instead of “What was the previous word?”, the model asks **“Which words matter for this word?”**

Encoder–Decoder Architecture



Not every model uses both halves: BERT is encoder-only, GPT-style models are decoder-only.

Inside an Encoder



Encoder layer × N

Self-Attention
Relates every token to the others

Feed-Forward
Transforms each token’s representation

Residual connection
Preserves information from earlier layers

Layer normalization
Stabilizes training

The Attention Mechanism

Q **Query**
What am I looking for?

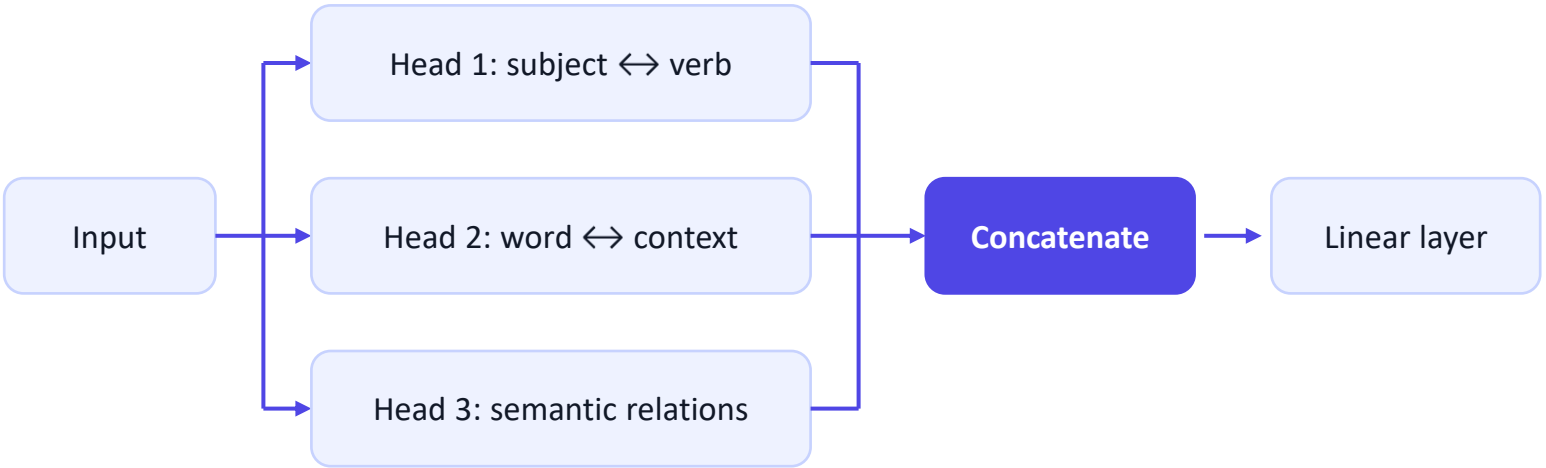
K **Key**
What does each token offer for matching?

V **Value**
What information is passed forward?

$$\text{Attention}(Q, K, V) = \text{softmax}(QK^T / \sqrt{d_k}) V$$

Similarity scores → Softmax → Attention weights → Weighted V → Contextual output

Multi-Head Attention



Different heads learn different relationships: grammar, meaning and context. Their outputs combine into a richer representation.

Other Key Components

1 Positional Encoding

Embedding + Position

Tokens run in parallel, so order is added: “Dog bites man” ≠ “Man bites dog”.

2 Feed-Forward Network

Linear → Activation → Linear

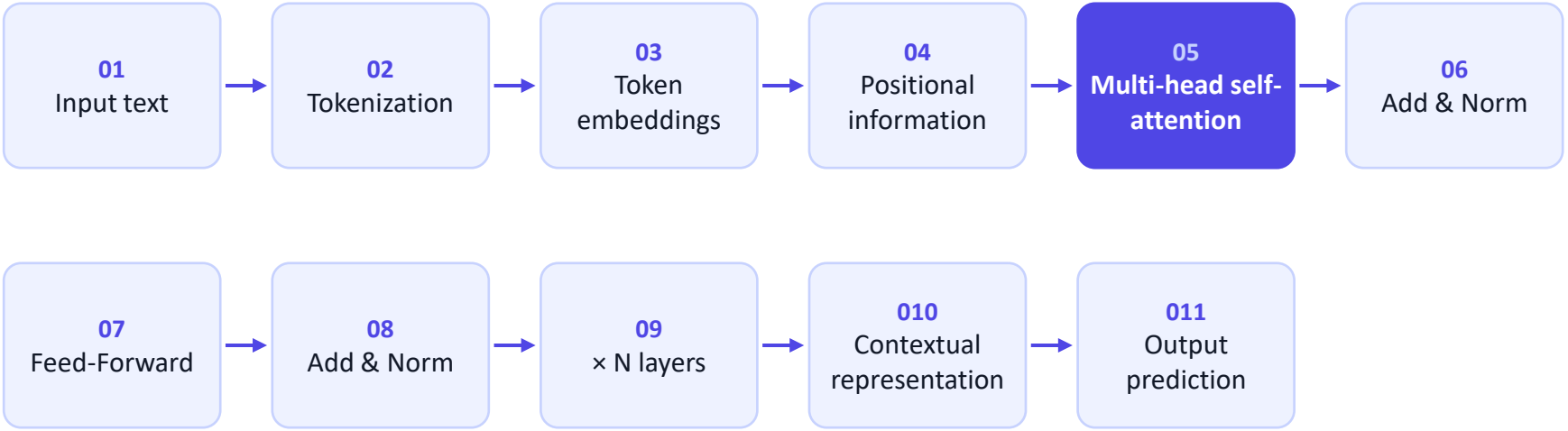
Transforms each token’s contextual representation.

3 Add & Norm

x + Sublayer(x) → Normalize

Add: residual connection keeps information. Norm: stabilizes training.

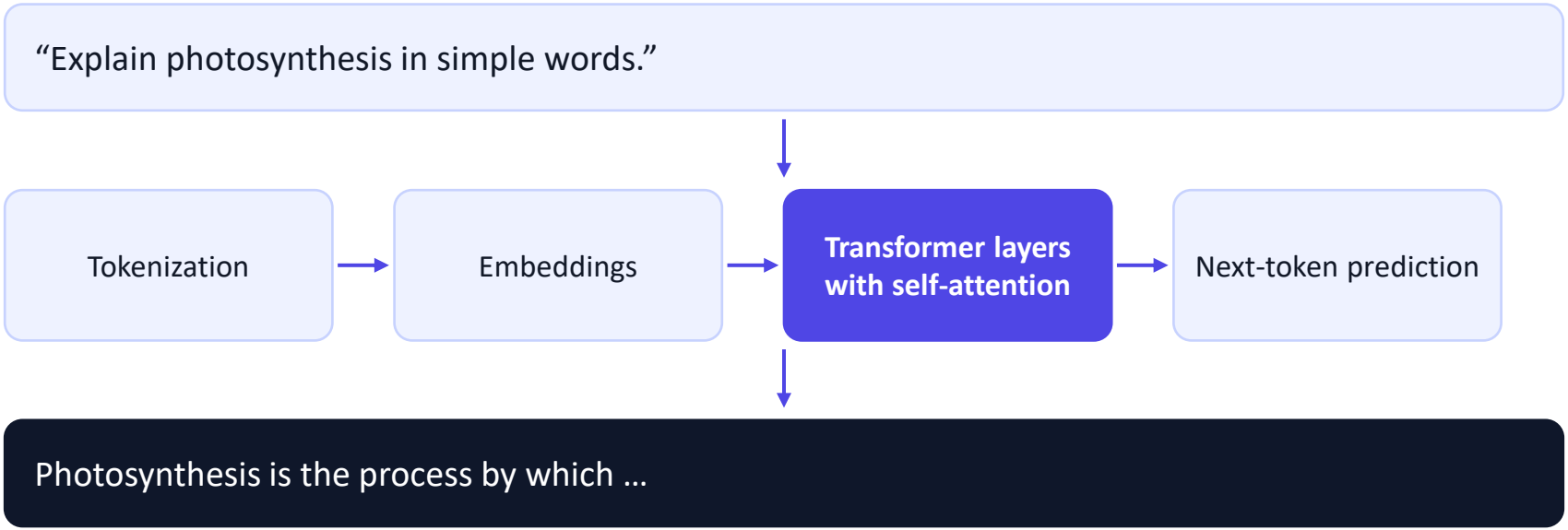
How a Transformer Works



The model repeatedly combines contextual information and transforms it until the representation suits the task.

REAL-WORLD EXAMPLE

Example: ChatGPT



The model does not retrieve a pre-written answer. It generates the response token by token.

REAL-WORLD EXAMPLE

How the Answer Is Generated

Explain | photosynthesis | in | simple | words

1. **Tokenize**
   Convert text into tokens
2. **Embed**
   Map tokens to numerical vectors
3. **Self-Attention**
   Relate tokens to each other
4. **Feed-Forward**
   Transform each representation
5. **Predict**
   Choose the next token
6. **Repeat**
   Until the response is complete

Photosynthesis → is → the → process → by → which → …

IMPACT

Advantages and Applications

Advantages

- Parallel processing in training
- Long-range dependencies
- Highly scalable
- Works with very large datasets
- Flexible architecture
- Rich contextual representations

Applications

Natural Language Processing
Translation, text generation, question answering, summarization, sentiment analysis

Other domains
Computer vision, speech, code generation, multimodal AI, recommendation systems

Examples BERT | GPT | T5 | ViT | Multimodal models

KEY TAKEAWAYS

Conclusion

“The key innovation of the Transformer is not simply processing more data. It is learning what information matters and how different parts of the input relate through Attention.”

1. Attention replaced recurrence
2. Built to overcome RNN and LSTM limits
3. Self- and multi-head attention capture relationships
4. Feed-forward, residual and norm layers refine and stabilize
5. Foundation of modern NLP, vision, speech and multimodal AI